

DDL: May 24th

9.7b

7.1(b) For the equilibrium, $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$, the degree of dissociation, α , at 298 K is 0.201 at 1.00 bar total pressure. Calculate (a) $\Delta_r G$, (2) K , and (3) $\Delta_r G^\circ$ at 298 K.

9.8a

7.2(a) Dinitrogen tetroxide is 18.46 per cent dissociated at 25°C and 1.00 bar in the equilibrium $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$. Calculate (a) K at 25°C, (b) $\Delta_r G^\circ$, (c) K at 100°C given that $\Delta_r H^\circ = +57.2 \text{ kJ mol}^{-1}$ over the temperature range.

9.9a

7.3(a) From information in the *Data section*, calculate the standard Gibbs energy and the equilibrium constant at (a) 298 K and (b) 400 K for the reaction $\text{PbO}(\text{s}) + \text{CO}(\text{g}) \rightleftharpoons \text{Pb}(\text{s}) + \text{CO}_2(\text{g})$. Assume that the reaction enthalpy is independent of temperature.

9.10b

7.4(a) In the gas-phase reaction $2 \text{A} + \text{B} \rightleftharpoons 3 \text{C} + 2 \text{D}$, it was found that, when 1.00 mol A, 2.00 mol B, and 1.00 mol D were mixed and allowed to come to equilibrium at 25°C, the resulting mixture contained 0.90 mol C at a total pressure of 1.00 bar. Calculate (a) the mole fractions of each species at equilibrium, (b) K_x , (c) K , and (d) $\Delta_r G^\circ$.

9.11 b

7.5(b) The standard enthalpy of a certain reaction is approximately constant at $+125 \text{ kJ mol}^{-1}$ from 800 K up to 1500 K. The standard reaction Gibbs energy is $+22 \text{ kJ mol}^{-1}$ at 1120 K. Estimate the temperature at which the equilibrium constant becomes greater than 1.

9.16a

9.16 (a) Use the data in Table 2.6 of the *Data section* to decide which of the following reactions have $K > 1$ at 298 K.

(a) $\text{HCl}(\text{g}) + \text{NH}_3(\text{g}) \rightleftharpoons \text{NH}_4\text{Cl}(\text{s})$

(b) $2 \text{Al}_2\text{O}_3(\text{s}) + 3 \text{Si}(\text{s}) \rightleftharpoons 3 \text{SiO}_2(\text{s}) + 4 \text{Al}(\text{s})$

(c) $\text{Fe}(\text{s}) + \text{H}_2\text{S}(\text{g}) \rightleftharpoons \text{FeS}(\text{s}) + \text{H}_2(\text{g})$

9.19b

7.11(b) The dissociation vapour pressure of NH_4Cl at 427°C is 608 kPa but at 459°C it has risen to 1115 kPa. Calculate (a) the equilibrium constant, (b) the standard reaction Gibbs energy, (c) the standard enthalpy, (d) the standard entropy of dissociation, all at 427°C . Assume that the vapour behaves as a perfect gas and that ΔH° and ΔS° are independent of temperature in the range given.